

Selecting SLMs

Selection of SLMs is always based on application. For the sake of simplicity, we categorise applications as general-purpose and special. Although this is a very crude way of classification, it simplifies the selection criteria to a great extent.

General-purpose. This is when the purpose is to simply measure noise from various sources and log these instantaneously. Common applications are quantifying loudness of a source (industrial, community, traffic and so on). For example, to check noise emitted from the water pump and verify if it's within acceptable limits. For such applications, most-widely-used SLMs are type 2 with A frequency weighing complying with IEC 61672-1 standard.

Special. Special application refers to noise monitoring for any of the following purposes:

- Industrial and environmental
- Occupational
- Community/neighborhood
- Research

For all the above applications, SLMs that can be used are type 1 or type 2 with A and C frequency weighing and Leq measurement.

For environmental noise pollution monitoring, Central Pollution Control Board of India recommends a type 1 instrument with integrating (Leq) feature and compliance with IEC 61672-1 (2002-05) class 1 standard.

For occupational/workplace noise monitoring, a type 2 SLM is sufficient for general-purpose noise surveys as per OSHA. The reason for this is that the human ear is subjected to noise-induced hearing loss at 4000Hz (typically from 4000Hz to 6000Hz) and a type 2 SLM easily covers this frequency range.

For community noise monitoring, a type 2 SLM is sufficient as decibel values are more important than frequencies. Certain guidelines enforced by national and international bodies decide the decibel limit for each zone such as The WHO Guidelines for Community Noise 1999.

For laboratory and research purpose such as to evaluate the efficiency of earmuffs or noise abatement procedures, type 1 SLM is must.

Rishabh Kumawat, director of Ambitech Systems Pvt Ltd, is currently working on air quality research to create awareness on the health hazards posed by air pollution. He is responsible for developing collaboration with foreign brands for promotions under Instrumex.

the noise dosimeter takes accurate and detailed measurements throughout the day. This type of device is particularly useful for measuring daily noise exposure of workers who move about a lot during their shifts and for those who work at night.

"Our SLM is built on internal standard 94dB/1kHz square-wave generator for calibrating certain internal amplifier circuits. An electric condenser microphone is used for high accuracy and long-term stability," says Vijay Nagori, CEO of Instrumex & Machinery Sales Corp.

Optimus, the latest generation of SLMs, combines the latest digital technology with ease of use, guarantees high performance with class 1 and class 2 SLMs, measures everything you need simultaneously to save time, is designed for basic, occupational and environmental

noise measurements, meets standards from around the world (including the UK, EU, OSHA, MSHA and ACGIH), type approved from PTB, LNE & Applus+ to IEC 61672-1:2002, is easy to upgrade at any time and carries a 15-year warranty.

SLMs applications

Accurate SLMs are expensive and not accessible to average people. On the other hand, smartphones that are widely available allow for various downloadable applications.

Many of these apps allow smartphones to be used as an SLM and provide a cost-effective solution to help determine noise levels in different environments.

Based on this availability, questions that are asked frequently are:

1. Can a smartphone SLM app be used as an informative substitute for measuring sound levels?
2. If such uses occur, will the accuracy be acceptable to substitute for measurements made with a dedicated and calibrated SLM?

The answer to the first question is yes and to the second question the answer should be, "Let us not get carried away at this time." An analogy might be: Can a pickup do the same job as a semi-tractor? To a limited extent, yes, in that it can haul products, but will have its limitations.

Smartphone SLM apps were selected based on their ability to measure occupationally relevancy criterion, as given below:

1. Ability to report un-weighted (C/Z/flat) or A weighted sound levels
2. Slow and fast response
3. Equivalent continuous sound level (Leq) or time-weighted average (TWA)

Kardous and Shaw acquired representative samples of popular smartphone apps. Of the 130 for iOS, ten (Adv Decibel Meter, Decibel Meter Pro, iSPL Pro, Noise Meter, NoiSee, Sound Level Meter, Sound-Meter, (Real) SPL Meter, SPL Pro and SPLnFFT) met the selection criteria.

A total of 62 Android apps were examined, and only four (SPL Meter, deciBel Pro (dBA), dB Sound



Fig. 3: Smartphones' SLM applications